MIL-U-46126(MR)
28 June 1968

MILITARY SPECIFICATION

URANIUM ALLOY FOR BODY SECTION CENTER OF AMMUNITION COMPONENTS

1. SCOPE

1.1 Scope. This specification covers the requirements for cast and extruded uranium alloy for body section center of ammunition components.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-11470(Ord) - Radiographic Inspection, Qualification of Equipment, Operators and Procedures.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 - Metals; Test Methods

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-453 - Inspection, Radiographic

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1395

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation: B311-58-Standard Method of Test for Density of Cemented Carbides
ASTM Designation: E8-66-Tension Testing of Metallic Materials
ASTM Designation: E9-67-Compression Testing of Metallic Materials

ASTM Designation: E23-66-Notched Bar Impact Testing of Metallic Materials

ASTM Method E120-66a-Chemical Analysis of Titanium and Titanium-Base Alloys

ASTM Method E120-66aT-Chemical Analysis of Titanium and Titanium-Base Alloys (Tentative)

ASTM Method E146-65-Chemical Analysis of Zirconium and Zirconium-Base Alloys

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia. Pennsylvania 19103).

3. REQUIREMENTS

3.1 Manufacture. The uranium alloy shall be manufactured using uranium melting stock of a purity which has been approved by the procuring activity, and such alloying elements as necessary to meet the requirements specified herein.

3.2 Chemical composition. Unless otherwise specified in the contract or order, the uranium alloy shall be: Uranium (91.5 percent)...molybdenum (8 percent)...titanium (0.5 percent), and the chemical composition shall comply with the range proposed by the contractor. The contractor shall submit a certified copy of the analysis of each extrusion to the procuring activity. The statement of analysis shall include all elements intentionally added as well as the amounts of elements outlined in table I.

3.3 Heat treatment. When material is heat treated to meet the applicable mechanical property requirements of this specification, the details of the heat treating procedure shall be provided by the contractor and shall be forwarded with each lot at the time of shipment. (see 6.1c).

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Table I. Maximum content for elements present as impurities

Element	Maximum	
	percent	ppm
Carbon	0.016	160
Iron	0.0115	115
Oxygen	0.010	100
Hydrogen	0.0004	4
Nitrogen	0.010	100

3.4 Mechanical properties.

3.4.1 Tension properties.

3.4.1.1 Tensile yield strength. Unless otherwise specified in the contract or order, the 0.1 percent offset tensile yield strength of each test specimen shall be 127,000 psi or greater.

3.4.1.2 Ductility. The average reduction of area shall be 25 percent or greater, and the average elongation shall be 9 percent or greater.

3.4.2 Compression properties.

3.4.2.1 Compressive yield strength. Unless otherwise specified in the contract or order, the 0.1 percent offset compressive yield strength of each test specimen shall be 125,000 psi or greater.

3.4.3 Impact resistance. The V-notch Charpy impact resistance (at -40°F ± 2°F) of each test specimen shall be 3.0 ft-lbs or greater.

3.5 Density. The density of each test specimen of the extruded material shall be 17.284 ± 0.100 grams per cubic centimeter.

3.6 Corrosion resistance. The corrosion specimens, when tested in accordance with 4.6.4, shall show no weight loss and a weight gain not to exceed 20 milligrams per square decimeter per day (mdd).

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3.7 Dimensions and dimensional tolerances. Dimensions and dimensional tolerances shall be as specified in the contract or order or on the applicable drawings.

3.8 Identification marking. Each piece shall be legibly and indelibly stencilled or painted with the number of this specification, manufacturer's identification, the piece number, condition, and heat number or designation. Indentation marking shall not be permitted. Each lot produced from melt stock supplied by the procuring activity shall be identified with the heat from which it is produced.

3.9 Soundness. The uranium material shall be free from laminations, inclusions, hard spots, cracks, and other injurious defects.

3.10 Workmanship. The uranium material shall be uniform in quality and condition, and shall be free from seams, folds, pits, scale, and other defects which might detrimentally affect the suitability of the material for the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all extrusions submitted for inspection at the same time, of the same heat, the same condition, the same diameter or thickness, and the same heat-treating and processing cycles.

4.3 Sampling.

4.3.1 For chemical analysis. At least one sample for chemical analysis shall be selected from each heat in accordance with method 111 or method 112 of Fed. Test Method Std. No. 151.

4.3.2 For mechanical properties tests. At least one sample shall be taken from each end of each body center section for mechanical properties tests.

4.4 Examination.

4.4.1 Visual. All material shall be subject to visual examination for compliance with identification marking (see 3.8) and workmanship requirements (see 3.10).

4.4.2 Dimensional. All material shall be subject to examination for compliance with dimensional requirements (see 3.7).

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4.4.3 Preparation for shipment. Examination shall be made to determine compliance with the requirements for the preparation for shipment (see section 5).

4.5 Tests.

4.5.1 Chemical analysis. Samples for chemical analysis shall be prepared and tested in accordance with method 111 or method 112 of Fed. Test Method Std. No. 151. In case of dispute, the analysis by method 111 shall be the basis for acceptance or rejection.

4.5.1.1 Determination of oxygen. The ASTM Method E120-66aT for determining oxygen by the Vacuum Fusion Method using the Platinum Flux or similar techniques, or any other applicable methods agreed upon by the contractor and procuring activity, shall be used in determining the percentages of oxygen in the uranium alloy.

4.5.1.2 Determination of hydrogen. The ASTM Method E146-65 for determining hydrogen by the Hot Extraction Method using an extraction temperature of 950°C, or any other applicable method agreed upon by the contractor and procuring activity, shall be used in determining the percentage of hydrogen in the uranium alloy.

4.5.1.3 Determination of nigrogen. The ASTM Method E120-66a, "Chemical Analysis of Titanium and Titanium-Base Alloys" or any other applicable method agreed upon by the contractor and procuring activity shall be used in determining the percentages of nitrogen in the uranium alloy.

4.5.1.4 Determination of carbon. Carbon content in the uranium alloy shall be determined by the High Frequency Furnace Combustion Method or any other applicable method agreed upon by the contractor and procuring activit (see 6.3).

4.5.2 Mechanical properties tests.

4.5.2.1 Location of test specimens. Unless otherwise specified in the contract or order, specimens for impact and tension tests shall be taken transverse to the direction of major working, and specimens for compression testing shall be taken in the longitudinal direction. The location of specimens shall be in accordance with figure 1.

4.5.2.2 Tension tests. At least two tensile test specimens shall be taken from each sample, and shall be prepared and tested in accordance with ASTM method E8-66. The yield strength shall be determined by the offset method by plotting a stress-strain diagram. The limiting offset shall be 0.10 percent (0.001 inch per inch of gage length). Unless otherwise approved by the procuring activity, the strain rate shall not exceed 0.005 in/in/min. up to the yield strength at 0.2 percent offset. The loads and strains shall be accurately indicated throughout the test.

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4.5.2.3 Compression tests. At least two compression test specimens shall be taken from each sample. The test specimens shall be approximately 3/8 inch in diameter and 3/4 inch in length. The compressive strength shall be determined by the offset method by plotting a stress-strain diagram. The limiting offset shall be 0.10 percent. Unless otherwise approved by the procuring activity, the strain rate shall not exceed 0.005 in/in/min. The compression test shall be conducted in accordance with ASTM Designation E9-67, Compression Testing of Metallic Materials.

4.5.2.4 Impact tests. At least two standard Charpy V-notch impact test specimens shall be taken from each sample and shall be prepared and tested in accordance with ASTM E23-66, Notched-Bar Impact Testing of Metallic Materials. The temperature of the test specimens at the time of fracture shall be -40°F ± 2°F. Charpy machines used in testing shall have been proof tested (calibrated) within one year prior to the time of inspection testing.

4.5.3 Density tests. At least two density test specimens shall be taken from each sample, and shall be prepared and tested in accordance with ASTM Designation: B311-58. A density test shall also be conducted on each body section center after rough machining.

4.5.4 Accelerated corrosion tests. When required, static corrosion tests shall be conducted as shown in 4.5.4.1 through 4.5.4.2 (see 6.5). Samples for this test may be taken from Charpy V-notch impact test specimens. The samples shall be surface ground to approximately the following dimensions: 0.25 decimeters (length), 0.10 decimeters (width), and 0.025 decimeters (thickness), and the surface area shall be determined.

4.5.4.1 Pretreatment and cleaning of test specimens. Test specimens shall be vapor-degreased with trichloroethylene and then pickled by immersing in concentrated (approximately 70 percent) nitric acid to remove surface film. The specimens shall be immediately rinsed in distilled water, then in acetone, and shall then be air-dried and placed in a dessicator.

4.5.4.2 Corrosion test procedure. After approximately 10 minutes, the test specimens shall be removed from the dessicator and weighed to 0.1 milligram and immersed completely in clean 1000 ml. Erlenmeyer-type flasks, each containing 400 ml. of distilled water and sufficient number of glass beads to prevent bumping of the specimens. The solution shall be brought to a boil within 10 minutes and kept at a boil throughout the remainder of the test, and the water level maintained by use of a Liebig-type condenser. Duplicate tests shall be conducted using separate flasks and shall be carried out under a pressure of 1 atmosphere for a 24-hour period. After 24 hours, the test specimens shall be removed from the boiling water, rinsed under flowing tap water, rinsed in distilled water and in acetone, air-dried, and placed in a dessicator for approximately 10 minutes before being re-weighed. The weight change shall be converted to milligrams per square decimeter per day.

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4.5.5 Soundness. The uranium material shall be examined radiographically (see 6.6) in accordance with MIL-STD-453 at 2-2T-level of sensitivity or better or the material shall be examined by a combination of radiographic and other nondestructive methods such as ultrasonic and electromagnetic methods according to procedures which have been approved by the procuring activity.

4.5.5.1 Number of views. When radiographic examination is being performed, at least six views (60 degrees apart) shall be taken on each piece to insure 100 percent radiographic examination.

4.5.5.2 Film density. When radiographic examination is being carried out, the film density shall be 1.4 to 2.8 for all the cross sections.

4.6 Retest and rejection. Retest and rejection of individual body sections shall be in accordance with the general section of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 Level C. Packing shall be in accordance with commercial practice adequate to insure acceptance and safe delivery by the carrier for the mode of transportation employed.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of MIL-STD-129.

5.4 Documents.

5.4.1 The information required by 3.3 shall be forwarded with the material at the time of delivery.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Chemical composition if necessary (see 3.2).
(c) When material is to be heat treated (see 3.3).
(d) When other tensile and compressive strength requirements are acceptable (see 3.4.1.1 and 3.4.2.1).
(e) Dimensions and dimensional tolerances (see 3.7).
(f) Name of inspecting agency when inspection shall be performed by other than contractor (see 4.1).
(g) When mechanical property specimens are to be taken in a different direction (see 4.5.2.1).
(h) When a different strain rate is allowable (see 4.5.2.2 and 4.5.2.3).
(i) When corrosion tests are required (see 3.6 and 4.5.4).
(j) When special marking-for-shipment requirements are necessary (see 5.3).
(k) Special Atomic Energy Commission and Interstate Commerce Commission regulations and requirements pertaining to the safety and methods involved in the preservation, packing, marking, and safeguarding of uranium alloy material covered by this specification.

6.2 The average impurity level of the uranium melt for the five major impurities are as follows:

Element	Average content (ppm)
C	75
H	10
O	20
N	30
Fe	100

The maximum content of all other impurities combined will be approximately 500 ppm.

6.3 The method referred to in 4.5.1.4 is shown in Information Bulletin No. T8, Part II, issued by the Metallurgical Advisory Committee on Titanium, on pp. 12-19.

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6.4 Information regarding comparison tests on Charpy impact machines employing Army Materials and Mechanics Research Center comparison specimens can be obtained from the local procurement district or from the Commanding General, Army Materials and Mechanics Research Center, Watertown, Massachusetts 02172.

6.5 The contractor will not ordinarily be required to perform corrosion tests on acceptable uranium alloys previously tested and reported in literature.

6.6 Radiographic examination. In addition to the procedures specified in MIL-STD-453, the following radiographic procedures are suggested:

6.6.1 Except as provided in 6.6.2 and 6.6.3, a 1,000 kilovolt peak (KVP) resonant transformer type of X-ray equipment with a source to part distance of 24 inches should be used. Type M, fine grain film encased in 0.10 inch back and front screens should be placed in close contact with the inside surface of the part. One quarter of an inch of lead backing is required in back of the film to prevent back scattering. The penetrameter is placed on the piece, and lead numbers and figures indicating the heat number are placed on the piece in such a position as to be clearly visible in the negative. Exposures are made at 1 million electron volts (mev), 3 milliamperes (ma) at a source to part distance of 24 inches as mentioned above.

6.6.2 When using 2 mev constant potential electrostatic generator equipment, a source to part distance of 36 inches is required. Exposures are made at 2 mev, 110 micromilliamperes (mma). The film shall be placed in intimate contact with the inside surface of the part to prevent any possible distortion or blurring. The section size of the part at the time of the radiographic examination is approximately 0.60 inch. The penetrameter thickness shall be no greater than 2 percent of the section size being radiographed. For section size of 0.60 inch, the use of 2-2T radiography is acceptable.

6.6.3 Other radiographic equipment which meets the above requirements will be acceptable provided approval is obtained from the procuring activity.

6.7 This specification covers the requirements of WAPD-373A, dated 26 February 1963, which has been cancelled.

Custodian:
Army - MR

Preparing activity:
Army - MR

Review activities:
Army - MU

Project No. 1395-A055

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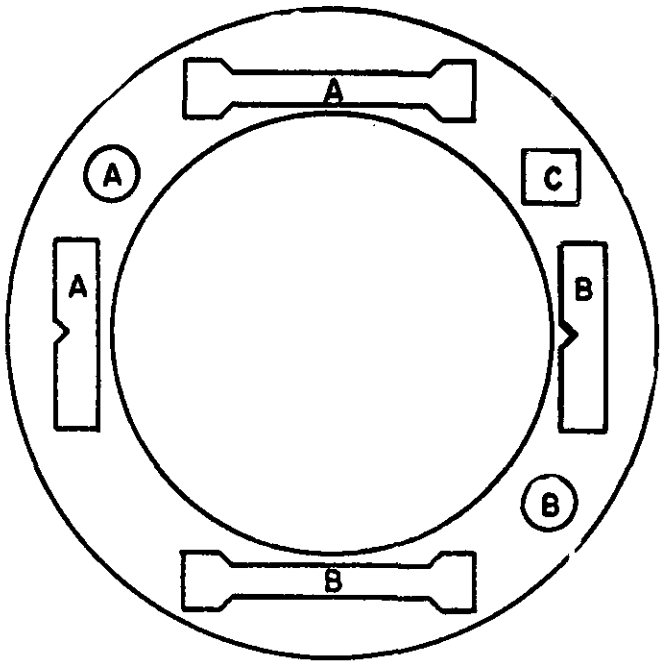


FIGURE I – LOCATION OF TENSILE, COMPRESSION, CHARPY AND DENSITY SPECIMENS

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	*QUANTITY OF ITEMS PROCURED*	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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